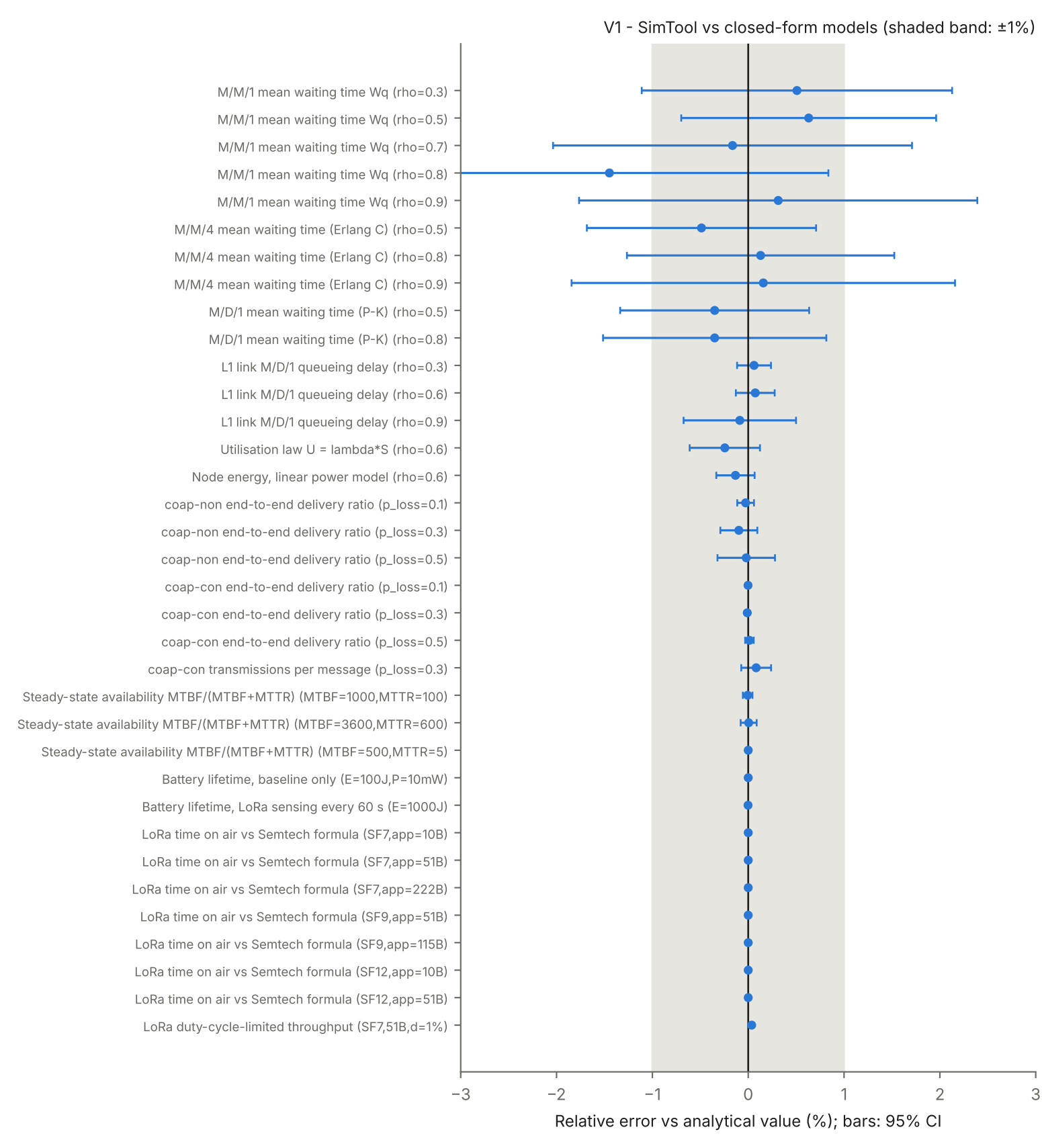
V1 - SimTool vs closed-form models (shaded band: ±1%)
M/M/1 mean waiting time Wq (rho=0.3)
M/M/1 mean waiting time Wq (rho=0.5)
M/M/1 mean waiting time Wq (rho=0.7)
M/M/1 mean waiting time Wq (rho=0.8)
M/M/1 mean waiting time Wq (rho=0.9)
M/M/4 mean waiting time (Erlang C) (rho=0.5)
M/M/4 mean waiting time (Erlang C) (rho=0.8)
M/M/4 mean waiting time (Erlang C) (rho=0.9)
M/D/1 mean waiting time (P-K) (rho=0.5)
M/D/1 mean waiting time (P-K) (rho=0.8)
L1 link M/D/1 queueing delay (rho=0.3)
L1 link M/D/1 queueing delay (rho=0.6)
L1 link M/D/1 queueing delay (rho=0.9)
Utilisation law U = lambda*S (rho=0.6)
Node energy, linear power model (rho=0.6)
coap-non end-to-end delivery ratio (p_loss=0.1)
coap-non end-to-end delivery ratio (p_loss=0.3)
coap-non end-to-end delivery ratio (p_loss=0.5)
coap-con end-to-end delivery ratio (p_loss=0.1)
coap-con end-to-end delivery ratio (p_loss=0.3)
coap-con end-to-end delivery ratio (p_loss=0.5)
coap-con transmissions per message (p_loss=0.3)
Steady-state availability MTBF/(MTBF+MTTR) (MTBF=1000,MTTR=100)
Steady-state availability MTBF/(MTBF+MTTR) (MTBF=3600,MTTR=600)
Steady-state availability MTBF/(MTBF+MTTR) (MTBF=500,MTTR=5)
Battery lifetime, baseline only (E=100J,P=10mW)
Battery lifetime, LoRa sensing every 60 s (E=1000J)
LoRa time on air vs Semtech formula (SF7,app=10B)
LoRa time on air vs Semtech formula (SF7,app=51B)
LoRa time on air vs Semtech formula (SF7,app=222B)
LoRa time on air vs Semtech formula (SF9,app=51B)
LoRa time on air vs Semtech formula (SF9,app=115B)
LoRa time on air vs Semtech formula (SF12,app=10B)
LoRa time on air vs Semtech formula (SF12,app=51B)
LoRa duty-cycle-limited throughput (SF7,51B,d=1%)
−3
−2
−1
0
1
2
3
Relative error vs analytical value (%); bars: 95% CI